

Finite fields

Corollary

k : finite field of
characteristic p

$$(\mathbb{F}_p \leq k)$$

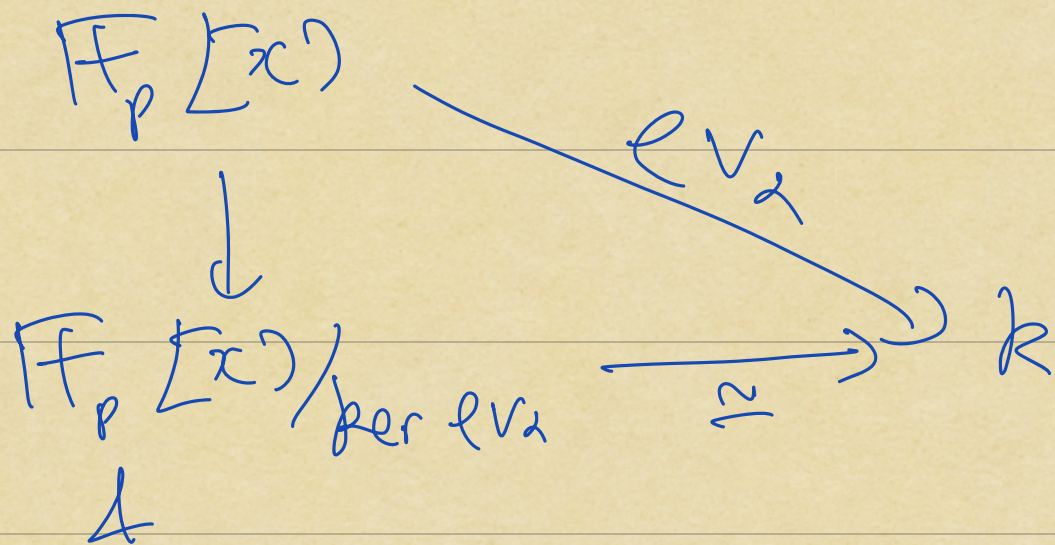
Then $\exists$ irreducible polynomial

$$f(x) \in \mathbb{F}_p[x] \text{ \& an}$$

isomorphism of fields

$$\frac{\mathbb{F}_p[x]}{(f(x))} \xrightarrow{\cong} k$$

Pf: $\alpha \in k^\times$: generator



$$\ker \text{ev}_\alpha = (f(x))$$

where $f(x) \in \mathbb{F}_p[x]$
is irreducible.

Thm (Galois)

For any prime p & any $d \geq 1$

there is up to isomorphism
a unique field of order
 p^d

Example: $p=3$, $d=2$

$$f_1(x) = x^2 + 1 \in \mathbb{F}_3[x]$$

$$f_2(x) = x^2 + x - 1$$

$$f_3(x) = x^2 - x - 1$$

$$\frac{\mathbb{F}_3[x]}{(x^2 + 1)}$$

$$\frac{\mathbb{F}_3[x]}{(x^2 + x - 1)}$$

$$\frac{\mathbb{F}_3[x]}{(x^2 - x - 1)}$$

$$\mathbb{F}_3[x]$$

$$\text{ev}_\alpha: \alpha^2 + 1 = 0$$

$$\frac{\mathbb{F}_3[x]}{(x^2 + 1)}$$

$$\frac{\overline{\text{ev}_\alpha}}{\cong}$$

$$\frac{\mathbb{F}_3[x]}{(x^2 + x - 1)}$$

$$(x^2 + x - 1)$$

$\mathbb{F}_3 \subseteq L$: field ext.

$$\text{Emb}_{\mathbb{F}_3} \left(\frac{\mathbb{F}_3[x]}{(x^2+1)}, L \right)$$

$$= \{ d \in L : d^2 + 1 = 0 \}$$

Concretely, the isomorphism $\overline{ev}_\alpha$ amounts to finding

$$d \in \frac{\mathbb{F}_3[x]}{(x^2+x-1)} \quad \text{s.t.} \quad d^2+1=0.$$

||

$$\left\{ \begin{array}{l} (x^2+x-1), \quad x + (x^2+x-1), \quad x+1 + (x^2+x-1) \\ 1 + (x^2+x-1), \quad (x-1) + (x^2+x-1), \quad (-x-1) + (x^2+x-1) \\ -1 + (x^2+x-1), \quad (-x+1) + (x^2+x-1), \quad (-x) + (x^2+x-1) \end{array} \right\}$$

$$((x-1)+1)^2 + [(x-1)+1] - 1 = 0$$

$$d = x-1$$

$$\Rightarrow (d+1)^2 + [d+1] - 1 = 0$$

$$\Rightarrow d^2 + 2d + 1 + d + 1 - 1 = 0$$

$$\Rightarrow x^2 + 1 = 0.$$

Rmk (Concrete interpretation
of Galois' theorem)

(1) (Existence)

$\forall d \geq 1, \exists$ an irred. poly.

$f(x) \in \mathbb{F}_p[x]$
of degree d

$\left[\frac{\mathbb{F}_p[x]}{(f(x))} \right]$ will be a field of
order p^d

(2) (Uniqueness)

If K is a field of
order p^d , then K
contains a zero for

every irreducible polynomial

$f(x) \in \mathbb{F}_p[x]$ of deg. d .

This will give an
isom. $\mathbb{F}_p[x]/(f(x)) \xrightarrow{\cong} k$
Where $\alpha \in k$ is a zero for
 $f(x)$

Fact (1) : $|k| = p^d$

$$\forall \alpha \in k, \alpha^{p^d} - \alpha = 0$$

i.e. Every element of
 k is a zero for $x^{p^d} - x$

Fact (2) :

$$x^{p^d} - x = \prod_{f(x) \in \text{Irr}_{1d}(\mathbb{F}_p)} f(x)$$

where $\text{Irr}_{1d}(\mathbb{F}_p) = \left\{ f(x) \in \mathbb{F}_p[x] : \right.$
"divides d "

- $f(x)$ irreducible
- has leading coeff 1

 $\left. \right\}$

$$(\cdot \deg f(x) | d)$$

Rem: We know that

$x^{\text{nd}} - x$ is a product
of irreducible polynomials
with leading coeff 1 &
that this factorization
is unique up to permutation

Note: Unique factorization
depends on the field of
Coefficients

Example: $\mathbb{R} \subseteq \mathbb{C}$

$$x^2 + 1 \in \mathbb{R}[x], \quad x^2 + 1 \stackrel{\text{unique factor}}{=} x^2 + 1 \in \mathbb{R}[x]$$

But $x^2+1 = (x+i)(x-i)$
 $\in \mathbb{Q}[x]$

Fact ② specialized to $p=3, d=2$

$$\underbrace{x^3 - x}_{\deg = 3} = \prod_{f(x) \in \text{Irr}_{12}(\mathbb{F}_3)} f(x)$$

$$= x \cdot (x-1) \cdot (x+1) \cdot \underbrace{\prod_{f(x) \in \text{Irr}_2(\mathbb{F}_3)} f(x)}_{\deg = 6}$$